

PATHOLOGY OF BONE VI

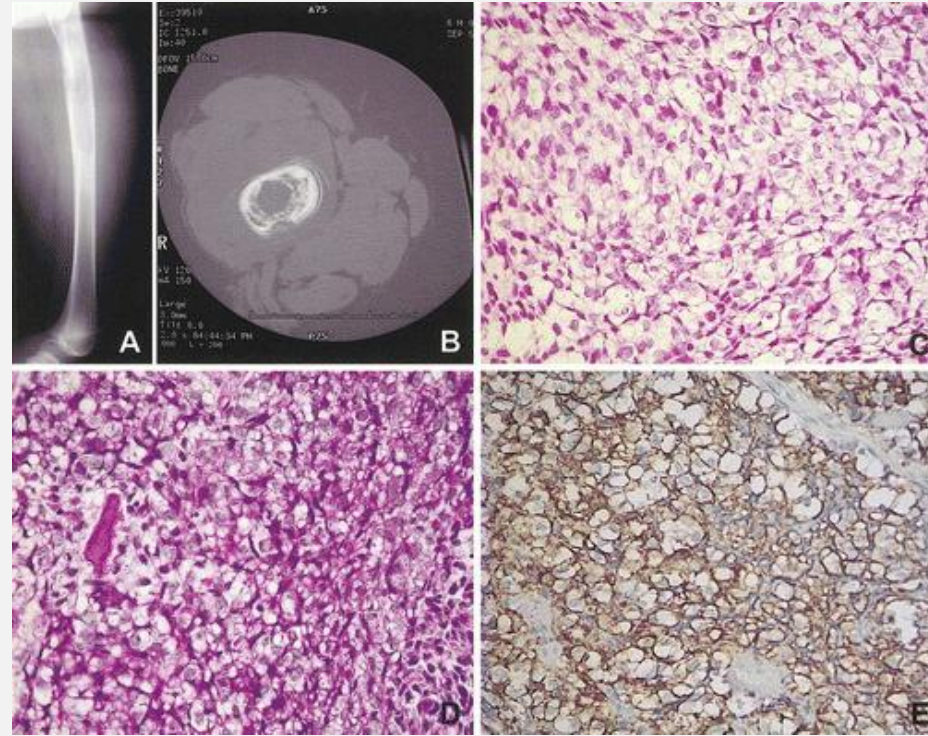
TUMORS ARISING FROM UNDIFFERENTIATED CELLS

I. **Ewing Sarcoma**

Case 7

A **19-year-old healthy** male presented with a 2-week history of pain in his right posterior thigh. The **pain** originated in the patient's right buttock and radiated to the back of the knee. Motor strength, sensory function and reflexes were normal. Musculoskeletal examination was within normal limits without tenderness of the hips or back.

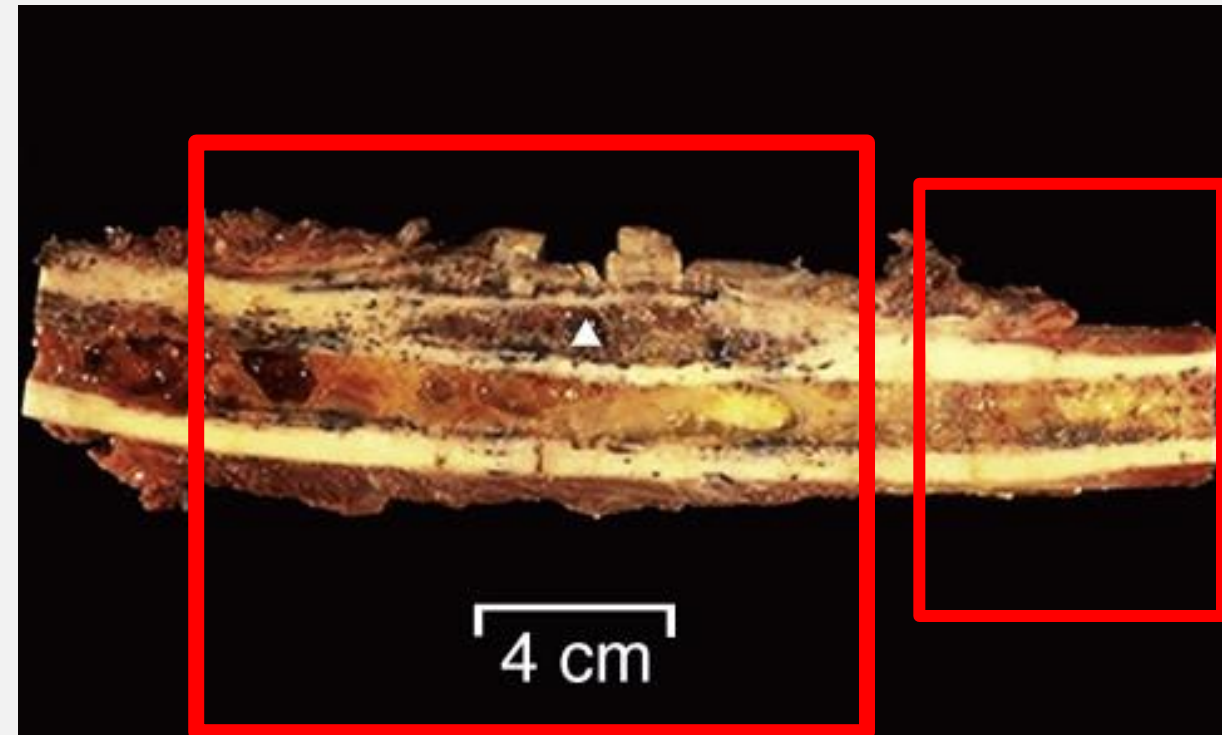
Despite the use of non-steroidal anti-inflammatory medications and physical therapy the **pain persisted**. Lab tests showed **leucytosis and elevated ESR**.



Gross morphology

- Arising in the **medullary cavity**
- Invading the cortex, periosteum, and soft tissue.
- The tumor is soft, *tan-white*, and frequently contains areas of *hemorrhage* and *necrosis*.

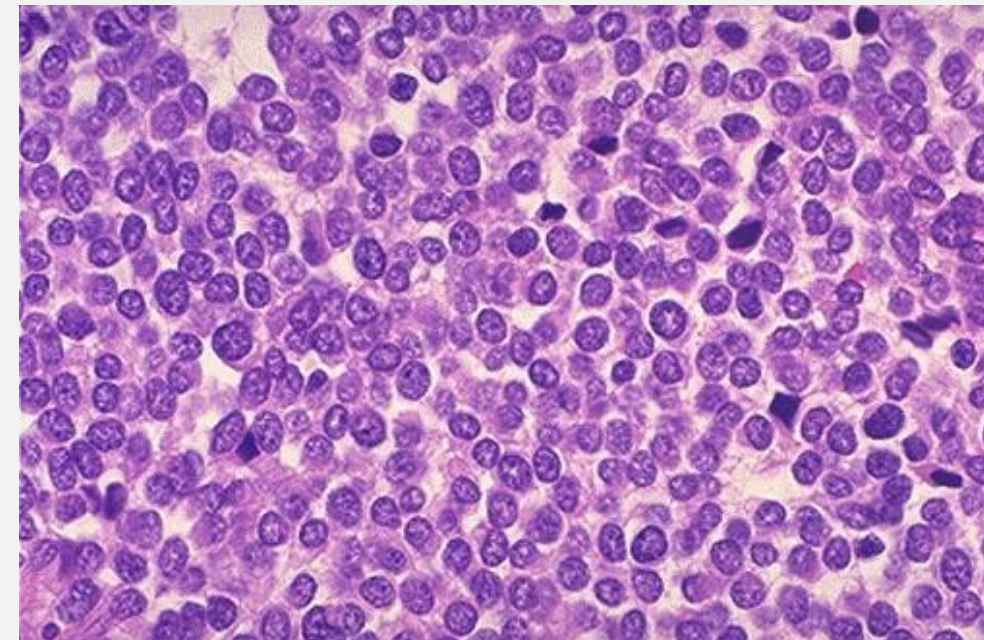
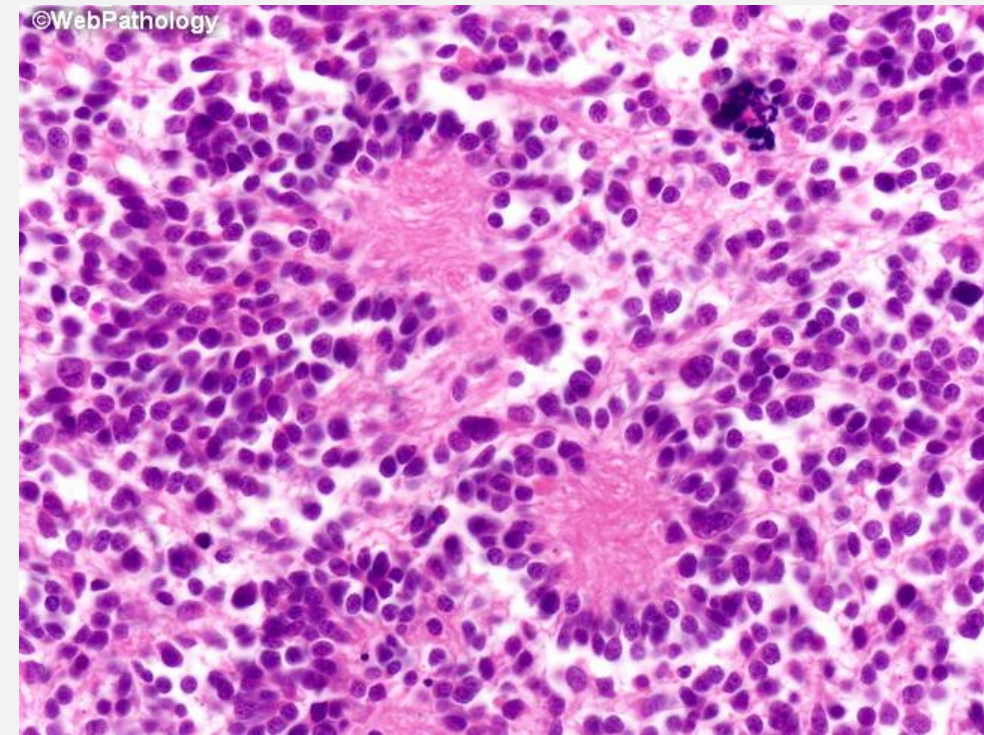
The fibular tumor mass shown here is breaking through (▲) the cortex. The tan tumor tissue has prominent areas of reddish hemorrhage and brownish necrosis. More normal long bone fatty marrow appears at the far right



Microscopic

- **small, round blue cell.**
- scant cytoplasm, which may appear clear because it is rich in glycogen.
- **Homer-Wright rosettes** (*round groupings of cells with a central fibrillary core*).
- The tumor cells **do not produce bone or cartilage.**

sheets of uniform small, round cells that are slightly larger and more cohesive than lymphocytes



Radiology

Destructive, **lytic tumor** with reactive periosteal bone
resembling **onion skin**



Ewing Sarcoma

- second most common primary malignant bone tumor in children
- adolescents (5-20) Males>females
- highly metastatic class of sarcoma
- Malignant neoplasm of **undifferentiated** cells
- Translocation of t(11:22)

Ewing Sarcoma

- Site: Femur and flat bones of pelvis
- Arise in **medullary cavity**
- May present with pain, fever, weight loss, leukocytosis and **increased erythrocyte sedimentation rate**
mimicking osteomyelitis

Ewing Sarcoma

- Ewing sarcoma is composed of small round cells with an increased nuclear-cytoplasmic ratio
- represents a **family of small round blue cell tumors** of childhood
 - retinoblastoma,
 - neuroblastoma,
 - rhabdomyosarcoma,
 - nephroblastoma

Treatment

This disease is **very aggressive**,
Common sites of metastases are lung, bones, and bone marrow.

Systemic chemotherapy is the mainstay of therapy, often being used before surgery.
Local treatment for the primary tumor includes **surgical resection**

Ewing's sarcoma at first presentation is a **curable tumor**, even in the presence of obvious metastatic disease, especially in children <11 years old.

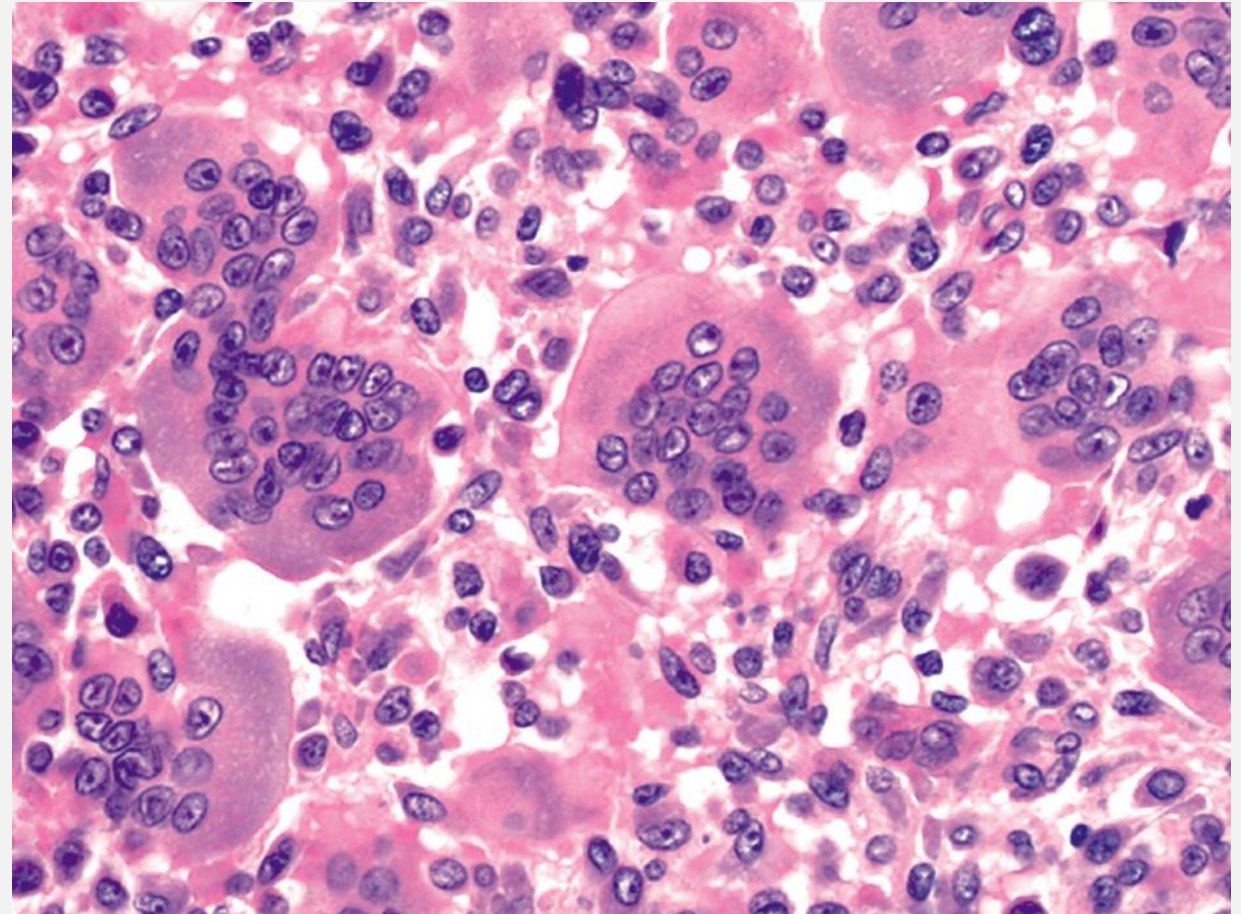
MISCELLANEOUS TUMORS



1. Giant cell tumor (osteoclastoma)
2. Fibrous dysplasia
3. Metastatic tumor

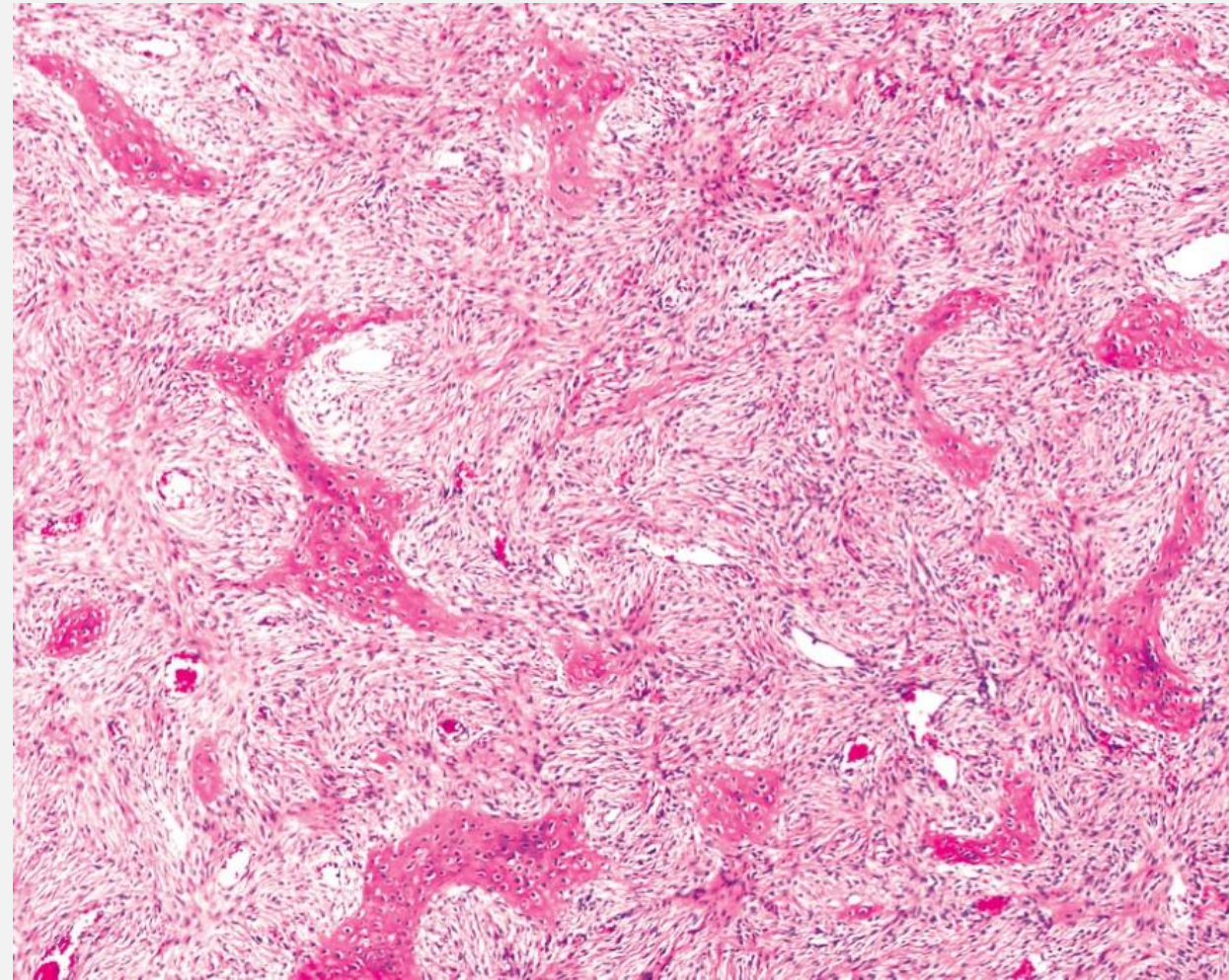
GIANT CELL TUMOR

- Also called **osteoclastoma**
- Benign but locally aggressive neoplasm with large numbers of **osteoclast-like giant cells**
- Associated with **Paget disease of bone**



FIBROUS DYSPLASIA

- Developmental, **non-neoplastic** disorder of bone-forming **mesenchyme**, causing **bone maturation arrest** at the woven bone stage



METASTATIC CANCER

- Most common malignant bone tumor is metastatic carcinoma
- **In adults,**
 - Prostate, Kidney, Lung, Thyroid and Breast **PKLT-B**
- **In children,**
 - Neuroblastoma, Wilm tumor,

METASTATIC CANCER

- Bone is the third most frequent site of metastasis, behind lung and liver.
- **Prostate and breast cancer** (BC) are responsible for the majority of the skeletal metastases (up to 70%).
- Bone metastasis are almost always **multiple and involve axial skeleton**

METASTATIC CANCER

- ***Osteolytic***: characterized by destruction of normal bone, present in multiple myeloma (MM), renal cell carcinoma, melanoma, non-small cell lung cancer, non-hodgkin lymphoma, thyroid cancer. The great majority of BC produces osteolytic metastases.
- ***Osteoblastic*** (or sclerotic), characterized by deposition of new bone, present in prostate cancer, carcinoid, small cell lung cancer, Hodgkin lymphoma or medulloblastoma.
- ***Mixed***, if a patient has both osteolytic and osteoblastic lesions, or if an individual metastasis has both osteolytic and osteoblastic components, present in BC, gastrointestinal cancers and squamous cancers.

Prostatic cancer mets to spine (sclerotic)

